

AWS WEEKLY TEST -01

Introduction to Amazon Web Service(AWS)

(AWS) is a cloud computing platform provided by Amazon. It offers on-demand IT services like servers, storage, databases, networking, and more — all delivered over the internet.

Instead of buying physical servers, companies can rent resources from AWS and pay only for what they use.

 Common AWS Services:

- EC2 (Elastic Compute Cloud) – Virtual servers
- S3 (Simple Storage Service) – Cloud storage
- RDS (Relational Database Service) – Managed databases
- Lambda – Serverless computing
- VPC (Virtual Private Cloud) – Networking service

 Advantages of AWS

 Cost Effective

- Pay-as-you-go pricing model
- No need to buy physical hardware  Scalability
- Easily increase or decrease resources
- Auto-scaling available  Global Infrastructure
- Data centers available worldwide  High Availability
- Services are reliable and fault-tolerant  Security
- Strong security features

- IAM (Identity and Access Management) for access control 膝

Large Service Ecosystem

- 200+ cloud services available
-

□ Disadvantages of AWS 膝

Complex Pricing

- Pricing structure can be confusing 膝 Learning Curve
- Many services → difficult for beginners 膝 Dependency on

Internet

- Requires stable internet connection 膝 Vendor Lock-in
- Difficult to migrate applications to another cloud provider 膏

Technical Support Cost

- Premium support plans are expensive

AWS Regions

An AWS Region is a physical geographic location in the world where AWS has multiple data centers.

Each region contains multiple Availability Zones (AZs).

Example AWS Regions:

- Mumbai (ap-south-1)
- US East (N. Virginia)
- London (eu-west-2)

- Singapore (ap-southeast-1)
-

🔗 What is Availability Zone (AZ)?

An Availability Zone is one or more data centers inside a region.

Example:

If you choose the Mumbai Region, it may have:

- ap-south-1a
- ap-south-1b
- ap-south-1c

Each AZ is isolated but connected to others for high availability.

🐛 Why Regions Are Important?

- Reduce latency (choose region close to users)
 - Data compliance requirements
 - Disaster recovery setup
 - High availability architecture
-

👉 Simple Example

If your users are in India, choose:

👣 Mumbai Region (ap-south-1)

If your users are in USA, choose:

👣 US East (N. Virginia)

This improves performance and reduces delay.

If you want, I can also explain:

- Difference between Region and Availability Zone

- AWS Global Infrastructure diagram explanation
- AWS Free Tier details
- AWS interview questions & answers 

AWS IAM Users and User Groups Creation.

What is IAM?

IAM lets you:

- Create Users (individual identities)
- Create Groups (collection of users)
- Assign Policies (permissions)
- Control access to AWS services securely

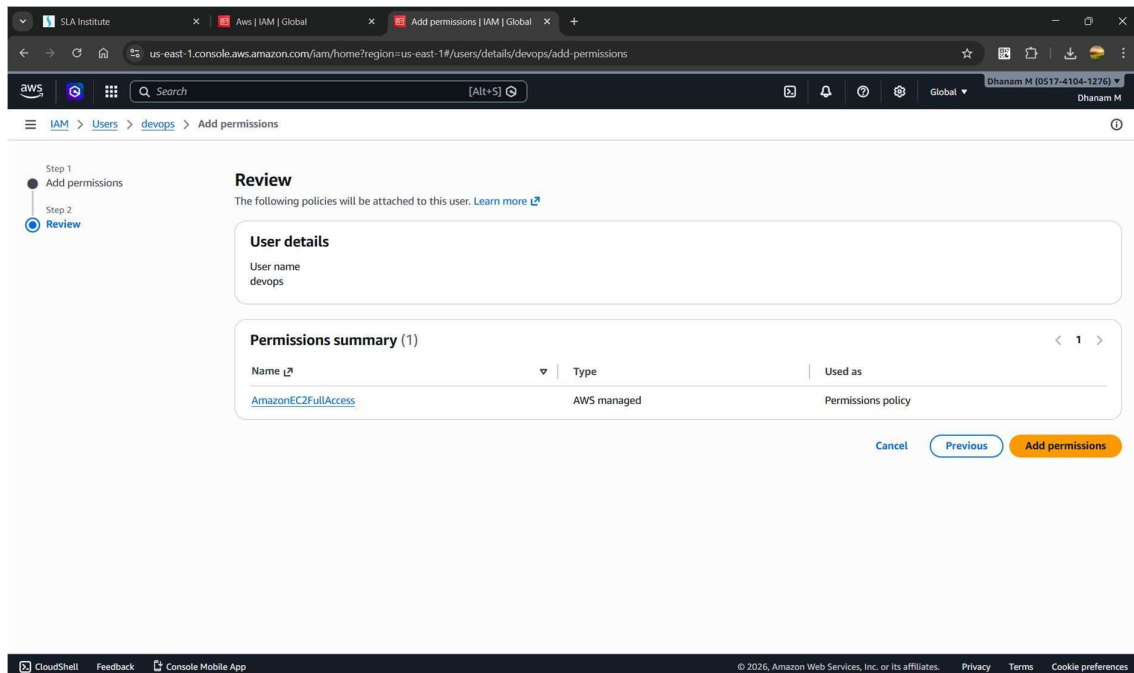
Step 1: specify the user details.

The screenshot shows the 'User details' configuration page in the AWS IAM console. It includes a 'User name' field with the value 'Devops1'. Below this is a checkbox for 'Provide user access to the AWS Management Console - optional', which is checked. Under 'Console password', there are two options: 'Autogenerated password' (selected) and 'Custom password'. A 'Show password' checkbox is also present. At the bottom, there is a blue information box stating: 'If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)'. Navigation buttons 'Cancel' and 'Next' are at the bottom right.

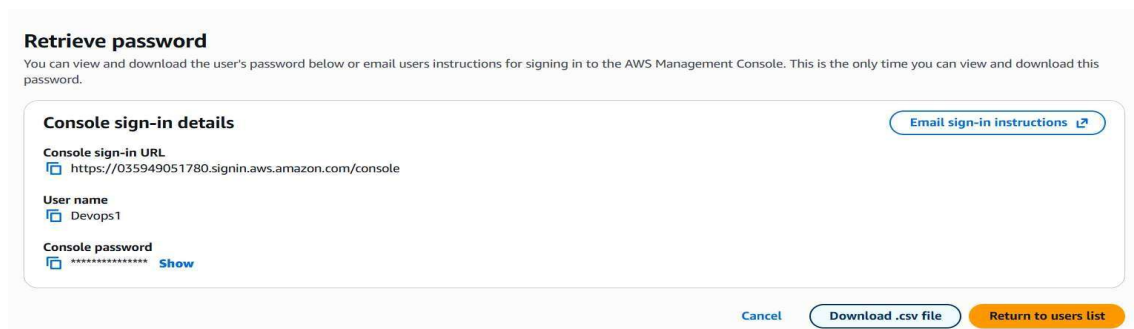
Step 2 : if you want you have to set permission.

The screenshot shows the 'Set permissions' configuration page in the AWS IAM console. It starts with the heading 'Set permissions' and a sub-heading 'Permissions options'. There are three radio button options: 'Add user to group' (selected), 'Copy permissions', and 'Attach policies directly'. Below these is a blue information box with the heading 'Get started with groups' and a 'Create group' button. At the bottom, there is a section for 'Set permissions boundary - optional' and navigation buttons 'Cancel', 'Previous', and 'Next'.

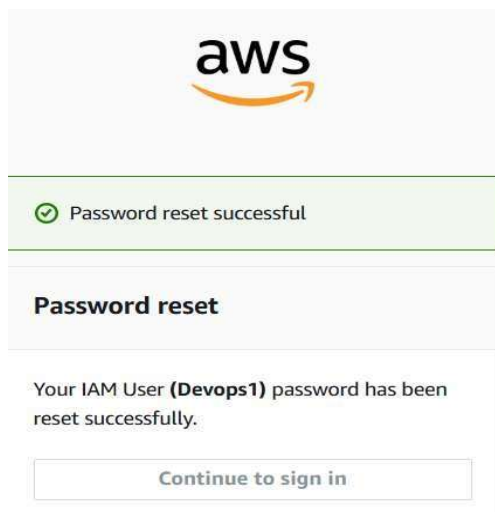
Step 3 : You can see the details of the permission and review the details.



Step 4 : you have to download the csv file and you have to mail it to the personnel.



Step 5 : then we have to create an new password for the personnel.



Creating an IAM Group

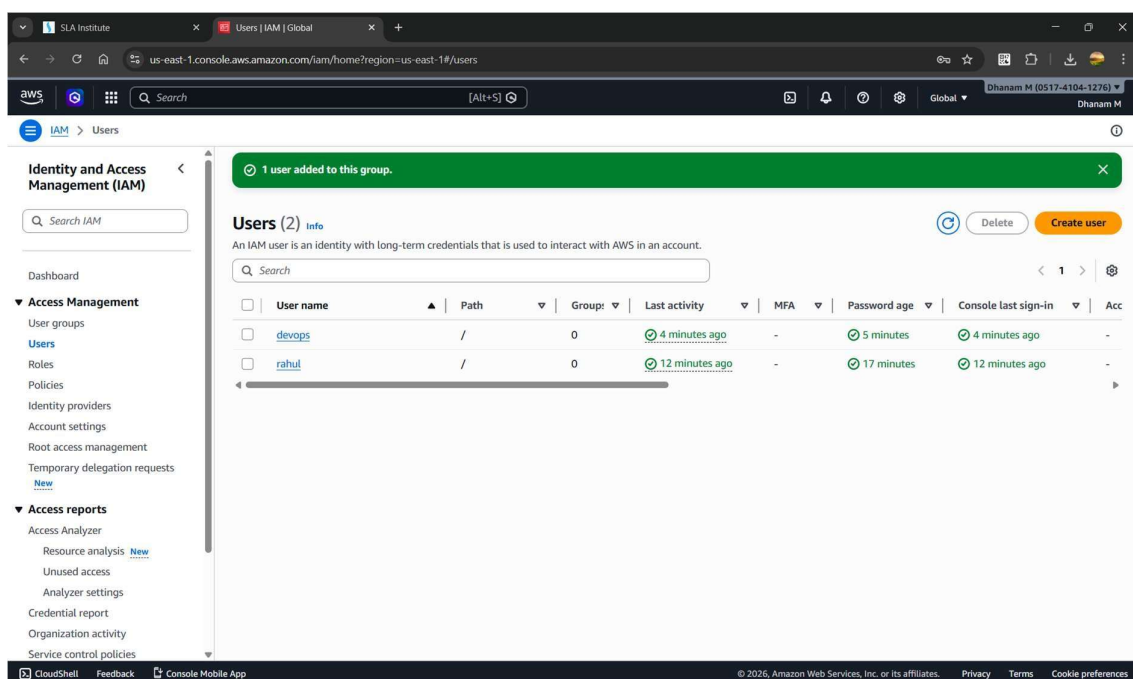
Groups are smart.

You don't assign permissions to users individually — you assign to groups.

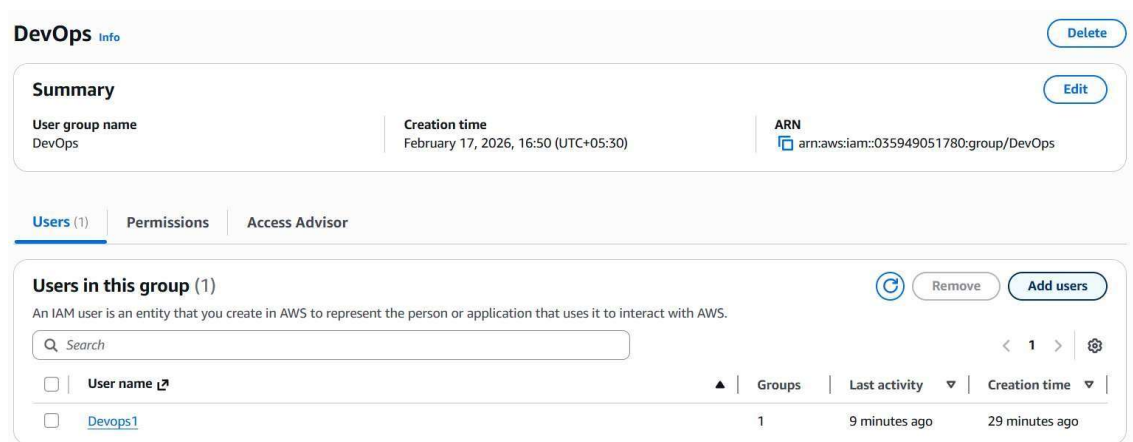
Example:

- Dev group → EC2 + S3 access
- Admin group → Full access
- ReadOnly group → View only

Step 1 : creating the user groups.



Step 2 : we can see the users in the group.



Add Users to Group

- Go to:
Users → Select user → Add to group → Select group → Save.

1 user added to this group.

Aws info [Delete](#) [Edit](#)

Summary

User group name: Aws | Creation time: February 17, 2026, 16:55 (UTC+05:30) | ARN: arn:aws:iam:051741041276:group/Aws

Users (2) | Permissions | Access Advisor

Users in this group (2) [Remove](#) [Add users](#)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

☐	User name i	Groups	Last activity	Creation time
☐	devops	1	3 minutes ago	5 minutes ago
☐	rahul	1	11 minutes ago	19 minutes ago

Giving permission to the user.

Step 1: Add permissions | Step 2: Review

Review

The following policies will be attached to this user. [Learn more](#)

User details

User name: Devops1

Permissions summary (1)

Name i	Type	Used as
AmazonEC2FullAccess	AWS managed	Permissions policy

[Cancel](#) [Previous](#) [Add permissions](#)

AWS IAM Roles and STS.

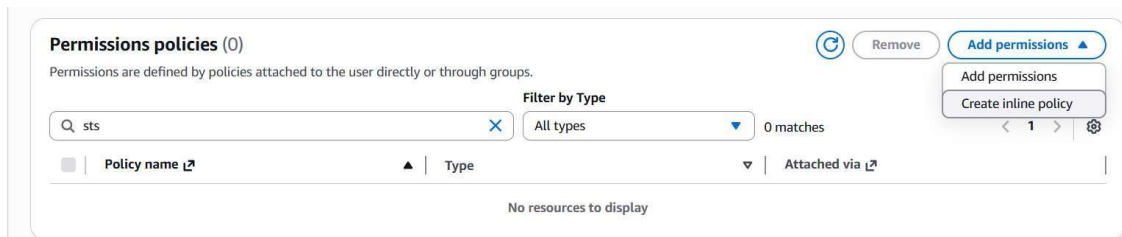
What is an IAM Role?

An IAM Role is a set of permissions that can be assumed temporarily.

Key point:

A role does NOT have permanent credentials. It gives temporary credentials via STS.

I have created an user and gave STS policy to that user.



This is the page of the STS.



You have to give all permission to the user policy.

Manual actions | [Add actions](#)

☒ All STS actions (sts:*)

Access level [Expand all](#) | [Collapse all](#)

- ▶ **Read** (Selected 5/5)
- ▶ **Write** (Selected 9/9)
- ▶ **Tagging** (Selected 2/2)

▼ **Resources**

Specify resource ARNs for these actions.

☒ All
☐ Specific

⚠ The all wildcard '*' may be overly permissive for the selected actions. Allowing specific ARNs for these service resources can improve security.

▶ **Request conditions - optional**

Actions on resources are allowed or denied only when these conditions are met.

Then I have created the IAM Role to give the access with the Roles.

And then I have connected using IAM User and Switched to the Role.

Switch Role

Switching roles enables you to manage resources across Amazon Web Services accounts using a single user. When you switch roles, you temporarily take on the permissions assigned to the new role. When you exit the role, you give up those permissions and get your original permissions back. [Learn more](#)

Account ID

The 12-digit account number or the alias of the account in which the role exists.

IAM role name

The name of the role that you want to assume which can be found at the end of the role's ARN. For example, provide the **TestRole** role name from the following role ARN: arn:aws:iam::123456789012:role/**TestRole**.

Display name - optional

This name will appear in the console navigation bar when active. Choose a name to help identify the permission set assigned to the role.

Display color - optional

The selected color displays in the console navigation when this role is active

☒ Red

Cancel

Switch Role

And then we will get this page.

It will give access to all the services.

Users | IAM | Global

us-east-1.console.aws.amazon.com/iam/home?region=ap-southeast-2#/users

Search [Alt+S]

Global

Dhanam M [0517-4104-1276]

developerrole @ 051741041276

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

- User groups
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests
- New

Access reports

- Access Analyzer
 - Resource analysis New
 - Unused access
 - Analyzer settings
- Credential report
- Organization activity
- Service control policies

Users (1) Info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

User name

Path

Group

Last activity

MFA

Password age

Console last sign-in

Acc

rahul

/

0

8 minutes ago

-

9 minutes

8 minutes ago

-

CloudShell

Feedback

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AWS creation of S3 bucket and usage.

What is S3?

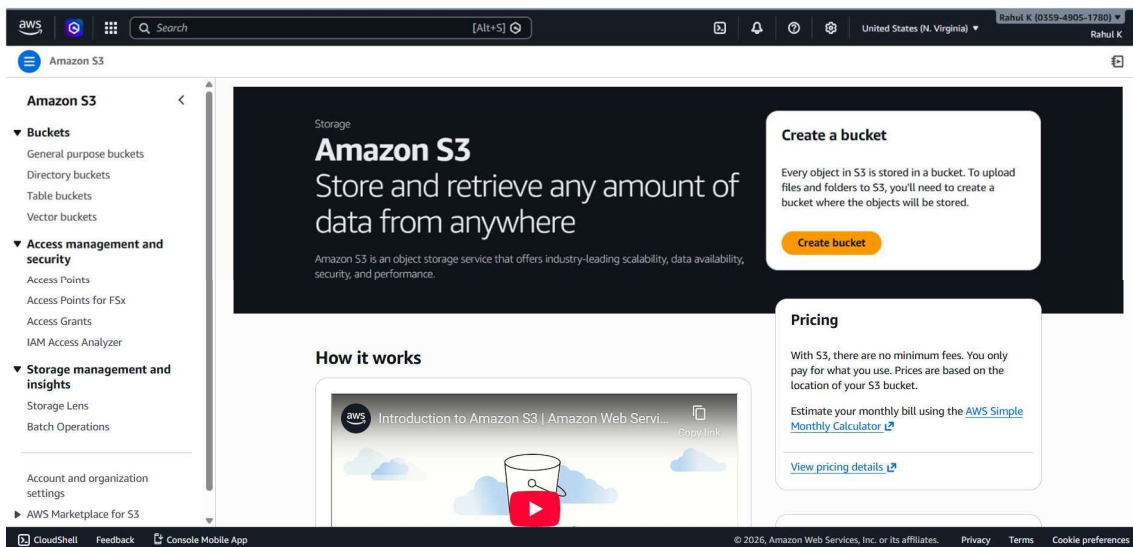
Amazon S3 (Simple Storage Service) is object storage.

It stores:

- Files
- Backups
- Static websites
- Docker artifacts
- Logs
- Terraform state files
- CI/CD build outputs

It is:

- Highly scalable ☐ 99.999999999% durable
- Pay for what you use



I am creating an new private S3 bucket .

Bucket type

Info

☒ General purpose
 Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ Directory
 Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name

Info

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

Format: s3://bucket/prefix

Object Ownership

Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ ACLs disabled (recommended)
 All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ ACLs enabled
 Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

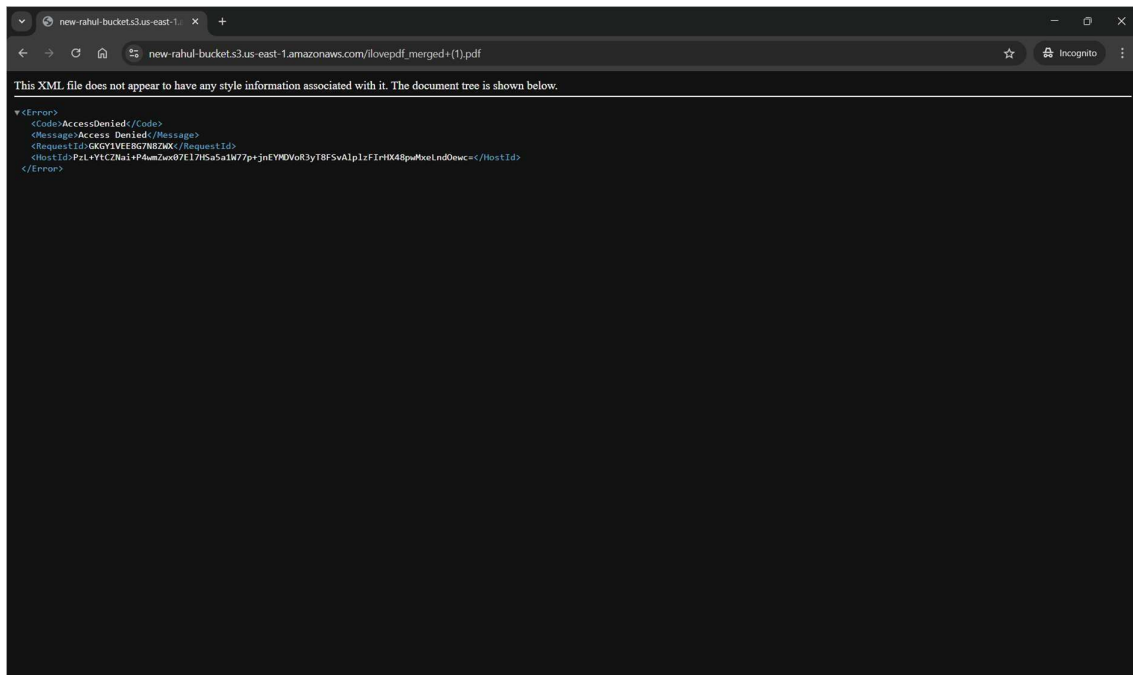
Object Ownership

Bucket owner enforced

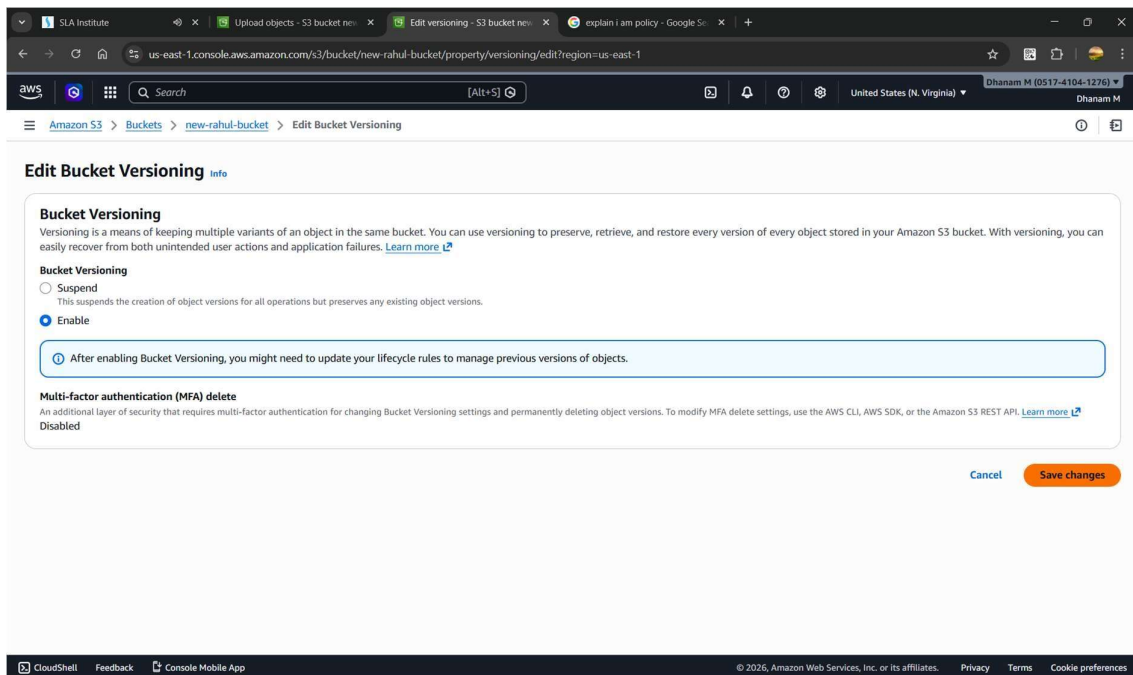
And then I have uploaded the files.

The screenshot displays the AWS Management Console interface for S3 Buckets. The main content area shows a list of 'General purpose buckets' with columns for Name, AWS Region, and Creation date. A single bucket, 'new-rahul-bucket', is listed in the 'us-east-1' region, created on February 19, 2026. The left-hand navigation pane includes sections for Buckets, Access management and security, and Storage management and insights. The right-hand pane features an 'Account snapshot' and an 'External access summary'.

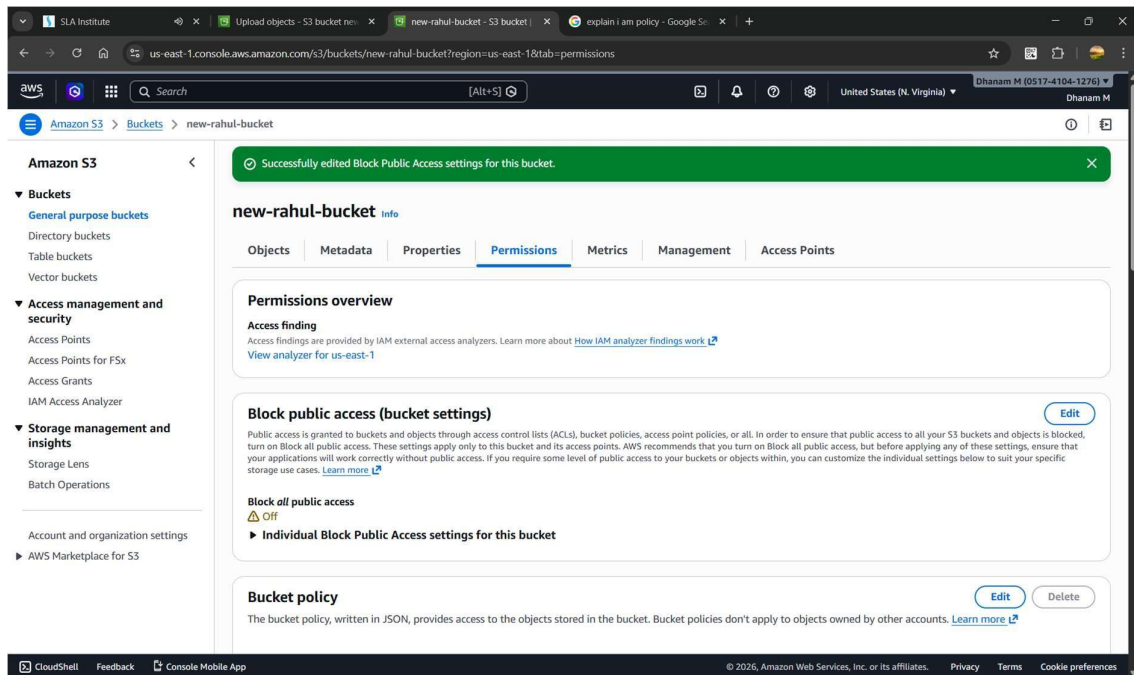
When I want to access the file it says that the access denied.



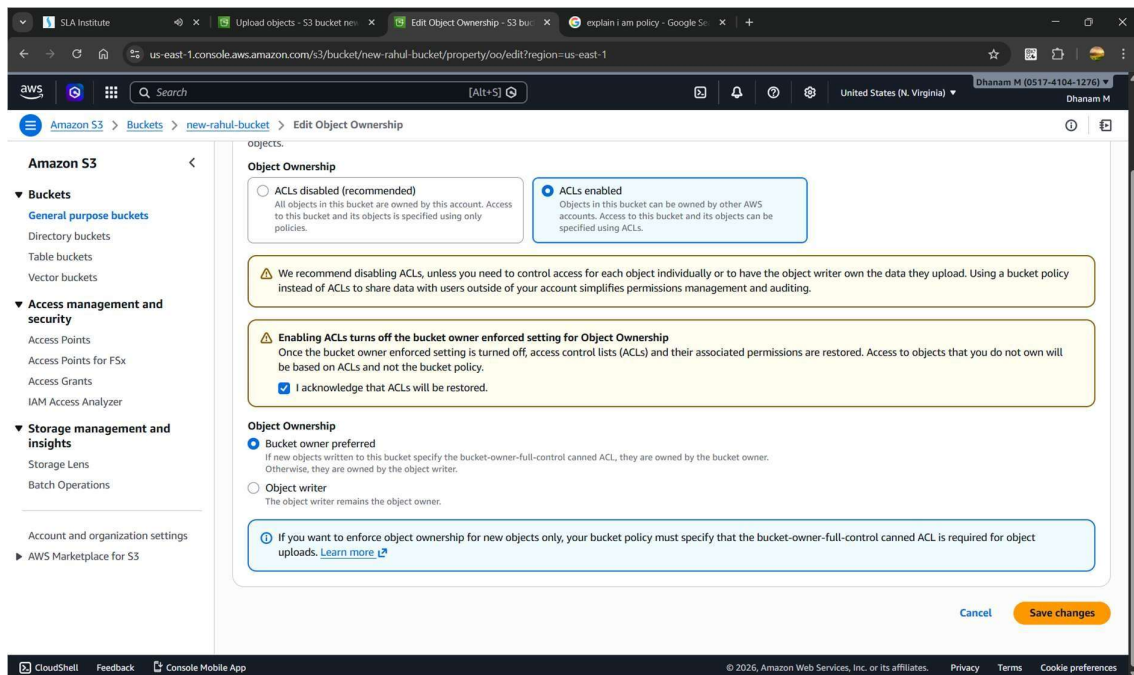
We have to enable versioning for the reusability of older versions.

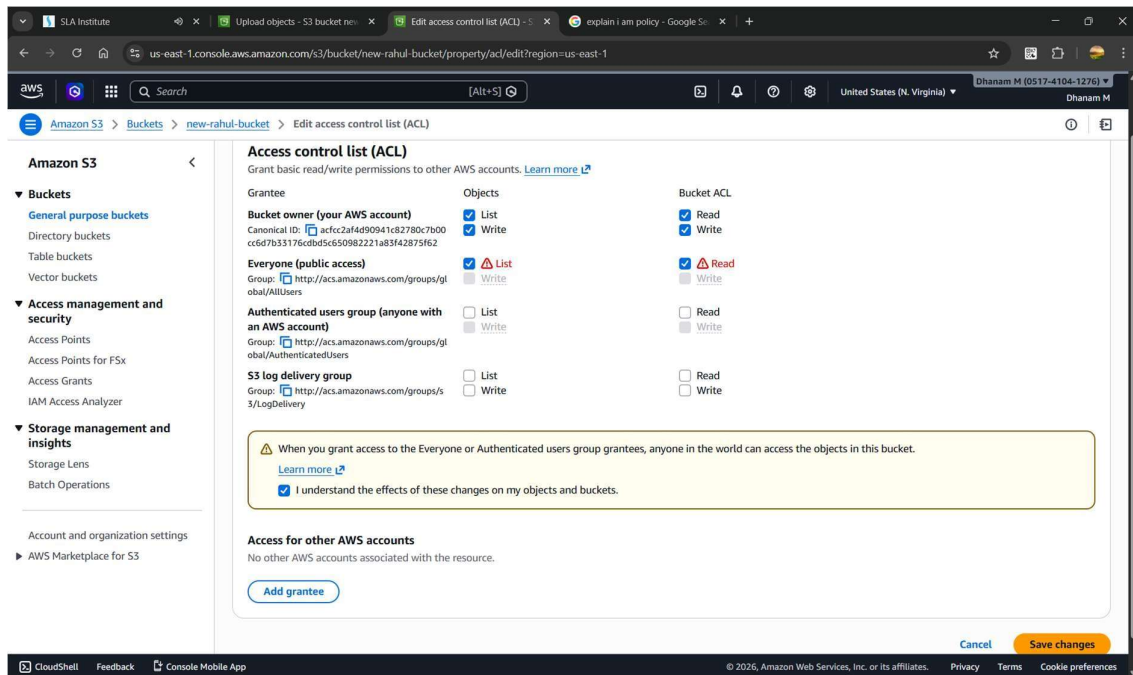


And we have to untick the all checkbox.

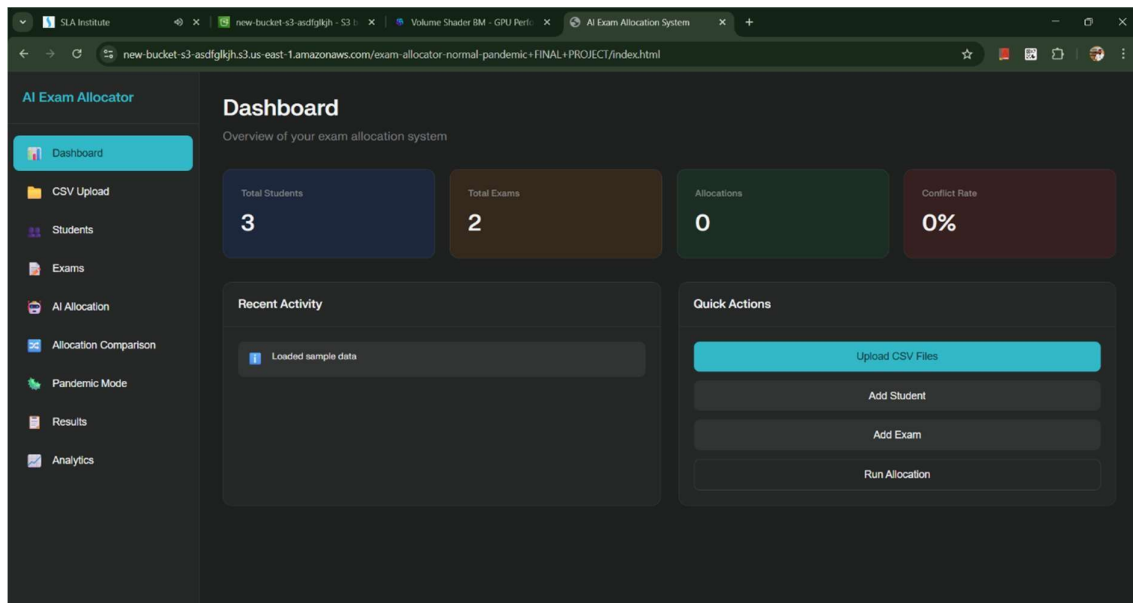


We have to give access to the user with ACL (Access Control Lists)





Then we will have the access to the website . S3 has an Adv. Of independent service so it act as an webserver also.



Bucket Versioning.

What it actually is

Bucket Versioning in Amazon S3 means every time you upload, modify, or delete a file (object), S3 keeps the old version instead of overwriting it.

No versioning = new upload replaces old file permanently.

With versioning = old file stays, new file gets a unique Version ID.

Why It Matters (Real Use Cases)

1. Accidental delete recovery ◦ If someone deletes project.zip, S3 just adds a Delete Marker. ◦ Old versions still exist. ◦ You can restore them.
2. Rollback capability ◦ Broke production config? ◦ Just restore previous version.
3. Protection from overwrites ◦ If someone uploads wrong data, you don't lose original.

Bucket versioning is only suspended not to be disabled , and I have enabled after creating the bucket.

Edit Bucket Versioning [Info](#)

Bucket Versioning
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
☐ Suspend
This suspends the creation of object versions for all operations but preserves any existing object versions.
☒ Enable

After enabling Bucket Versioning, you might need to update your lifecycle rules to manage previous versions of objects.

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)
Disabled

[Cancel](#) [Save changes](#)

Bucket versioning works like an commit id in github.

I have uploaded 3 files to the bucket.

Amazon S3

Upload: status [Close](#)

After you navigate away from this page, the following information is no longer available.

Summary

Destination s3://bucketversioning34565	Succeeded 3 files, 5.0 MB (100.00%)	Failed 0 files, 0 B (0%)
--	---	------------------------------------

Files and folders Configuration

Files and folders (3 total, 5.0 MB)
Find by name

Name	Folder	Type	Size	Status	Error
Day 2 Linux Navigation Comma...	-	application/pdf	1.4 MB	Succeeded	-
Day 3 admin and permission.pdf	-	application/pdf	1.9 MB	Succeeded	-
Day 4 networking.pdf	-	application/pdf	1.6 MB	Succeeded	-

The uploaded files.

bucketcersioning34565 Info

Objects (3)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix Show versions

☐	Name	Type	Version ID	Last modified	Size	Storage class
☐	Day 2 Linux Navigation Commands.pdf	pdf	GK5.ThDKPDRuwlwDvDg8ffCDU96GWEP	February 20, 2026, 17:14:43 (UTC+05:30)	1.4 MB	Standard
☐	Day 3 admin and permission.pdf	pdf	_JWtSleRJODXWismhd4ufegLw ah09CRJ	February 20, 2026, 17:14:44 (UTC+05:30)	1.9 MB	Standard
☐	Day 4 networking.pdf	pdf	.dJnErravNT4n3hwS27yal3SN5C rp_BH	February 20, 2026, 17:14:45 (UTC+05:30)	1.6 MB	Standard

The bucket versioning enabled means the bucket will create and version id we can retrieve the older version when we delete the file.

newbucket4622982 Info

Objects (7)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix Show versions

☐	Name	Type	Version ID	Last modified	Size	Storage class
☐	images/	Folder	-	-	-	-
☐	index.html	Delete marker	tLLP2l0jvt3fZlVnedCdkth3zL5p Gay	February 20, 2026, 17:04:34 (UTC+05:30)	0 B	-
☐	index.html	html	V7G9HfcUFXQSiqhLQVuiVTaC.L VQ8ZC	February 20, 2026, 16:57:58 (UTC+05:30)	18.9 KB	Standard
☐	templatemo-606-string-master.css	Delete marker	1S.JneELBL6c7PK1LJBPh_9SZe uWW6bZ	February 20, 2026, 17:04:34 (UTC+05:30)	0 B	-
☐	templatemo-606-string-master.css	css	1rqk73MnWN3cmQ6JJwT80Va OxNCqub0u	February 20, 2026, 16:57:58 (UTC+05:30)	23.1 KB	Standard
☐	templatemo-606-string-scripts.js	Delete marker	Bj5jjoOlx0zLxfWZyShZqeKNcFF yJwsa	February 20, 2026, 17:04:34 (UTC+05:30)	0 B	-
☐	templatemo-606-string-scripts.js	js	IcEn4rmDmIl4lwAq4e0Xgy.iOW cBiDOx	February 20, 2026, 16:57:59 (UTC+05:30)	13.2 KB	Standard

Bucket Replication.

What It Actually Is

S3 Bucket Replication automatically copies objects from one S3 bucket to another.

There are two types:

- CRR (Cross-Region Replication) → Different AWS regions
- SRR (Same-Region Replication) → Same region

Why Replication Exists

Disaster Recovery

- If your primary region fails, your data exists elsewhere.

Low Latency Access

- Users in another country? Store data closer to them.

Compliance

- Some laws require data to exist in specific regions.

Multi-Account Strategy

- Replicate from production account → backup account.

TYPES OF REPLICATION.

1. Cross-Region Replication (CRR)

Replicates objects to a different AWS region.

Example:

- Source → Mumbai region
- Destination → Singapore region

Used for:

- Disaster Recovery
- Global applications
- Region-level failure protection

Reality:

- Costs more (data transfer charges apply)
- Good for serious production setups

2. Same-Region Replication (SRR) Replicates

objects to another bucket in the same region.

Used for:

- Log copying

- Compliance separation
- Multi-team access segregation

Cheaper than CRR (no cross-region transfer cost)

3. Cross-Account Replication Replication between different AWS accounts.

Used for:

- Production account → Backup account
- Security isolation
- Enterprise setups Requires:
- IAM role permissions
- Bucket policy on destination

This is how mature DevOps teams operate.

4. Delete Marker Replication

Controls whether object deletions are replicated.

- Enabled → Delete in source = Delete in destination
- Disabled → Destination keeps object even if source deletes I have created two bucket in different regions.

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (3)

info

Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >

	Name	AWS Region	Creation date
☐	destinationbucket237664	Asia Pacific (Hyderabad) ap-south-2	February 20, 2026, 16:47:03 (UTC+05:30)
☐	newbucket4622982	US East (N. Virginia) us-east-1	February 20, 2026, 16:43:02 (UTC+05:30)
☐	sourcebucket1368	US East (N. Virginia) us-east-1	February 20, 2026, 16:44:59 (UTC+05:30)

Account snapshot

info

Updated daily

Storage Lens provides visibility into storage usage and activity trends.

View dashboard

External access summary

info

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

This the source bucket.

aws

Search

[Alt+S]

United States (N. Virginia)

Dhanam M (0517-4104-1276)

Amazon S3

Buckets

sourcebucket1368

sourcebucket1368

info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

Show versions

< 1 >

	Name	Type	Last modified	Size	Storage class
☐	Day 4 AWS creation of S3 bucket and usage.docx	docx	February 20, 2026, 16:52:39 (UTC+05:30)	891.6 KB	Standard

Create replication rule.

Amazon S3

Buckets

amazonsourcebucket3457

Replication rules

Create replication rule

Replication rule name

temp_rule

Up to 255 characters. In order to be able to use CloudWatch metrics to monitor the progress of your replication rule, the replication rule name must only contain English characters.

Status

Choose whether the rule will be enabled or disabled when created.

☒ Enabled
 ☐ Disabled

Priority

The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configuration at the highest priority and the priority can be changed on the replication rules table.

0

Giving the destination bucket details.

Amazon S3 > Buckets > amazonsourcebucket3457 > Replication rules > Create replication rule

☐ Limit the scope of this rule using one or more filters
☒ Apply to all objects in the bucket

Destination

Destination
 You can replicate objects across buckets in different AWS Regions (Cross-Region Replication) or you can replicate objects across buckets in the same AWS Region (Same-Region Replication). You can also specify a different bucket for each rule in the configuration. [Learn more](#) or [see Amazon S3 pricing](#)

☒ Choose a bucket in this account
☐ Specify a bucket in another account

Bucket name
 Choose the bucket that will receive replicated objects.
 [Browse S3](#)

Destination Region
 US East (N. Virginia) us-east-1

Create own IAM role.

Amazon S3 > Buckets > amazonsourcebucket3457 > Replication rules > Create replication rule

Destination Region
 US East (N. Virginia) us-east-1

IAM role
 Permission to access the specified resources
☒ Create new role
☐ Choose from existing IAM roles
☐ Enter IAM role ARN

If you want the existing data tick the yes option.

Replicate existing objects?

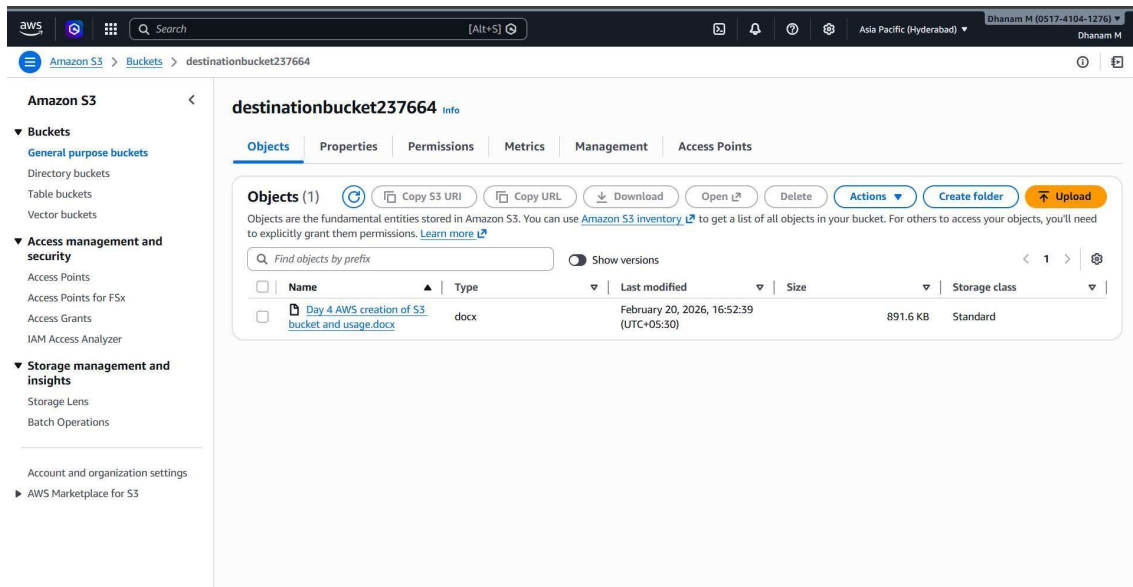
You can enable a one-time Batch Operations job from this replication configuration to replicate objects that already exist in the bucket and to synchronize the source and destination buckets. [Learn more](#) or [see pricing](#)

Existing objects

☐ No, do not replicate existing objects.
☒ Yes, replicate existing objects.

[Cancel](#) [Submit](#)

The destination bucket will be updated from 15 minutes to max hours.



Static Web Hosting.

What You Actually Get

- Cheap hosting
- High durability (11 9's)
- Auto scaling
- Public HTTP endpoint
- No server management

What you don't get:

- No server-side code execution
- No HTTPS by default (you need CloudFront for that)
- No database

To enable the static web hosting in the s3 we have to enable ACLs , Bucket versioning , and allow all public access (I have already done it).

aws

Search

[Alt+S] Ask Amazon Q

United States (N. Virginia)

Dhanam M

Amazon S3

Buckets

newbucket4622982

newbucket4622982

info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (4)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

Show versions

☐	Name	Type	Last modified	Size	Storage class
☐	images/	Folder	-	-	-
☐	index.html	html	February 20, 2026, 16:57:58 (UTC+05:30)	18.9 KB	Standard
☐	templatemo-606-string-master.css	css	February 20, 2026, 16:57:58 (UTC+05:30)	23.1 KB	Standard
☐	templatemo-606-string-scripts.js	js	February 20, 2026, 16:57:59 (UTC+05:30)	13.2 KB	Standard

ebhosting8745467

Successfully edited static website hosting.

Requester pays

Disabled

Static website hosting

Edit

Use this bucket to host a website or redirect requests. [Learn more](#)

We recommend using AWS Amplify Hosting for static website hosting

Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. [Learn more about Amplify Hosting](#) or [View your existing Amplify apps](#)

Create Amplify app

S3 static website hosting

Enabled

Hosting type

Bucket hosting

Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://staticwebhosting8745467.s3-website-us-east-1.amazonaws.com>

The hosted website is, using the given s3 url.

